Classification Assignment Grid

Problem Statement:

A requirement from the Hospital, Management asked us to create a predictive model which will predict the Chronic Kidney Disease (CKD) based on the several parameters. The Client has provided the dataset of the same.
1.) Identify your problem statement
2.) Tell basic info about the dataset (Total number of rows, columns)
3.) Mention the pre-processing method if you're doing any (like converting string to number – nominal data)
4.) Develop a good model with good evaluation metric. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.
5.) All the research values of each algorithm should be documented. (You can make tabulation or screenshot of the results.)
6.) Mention your final model, justify why u have chosen the same.

CLASSIFICATION ASSIGNMENT ANSWER

1.Problem Statement and understanding:

Client wants to predict whether a client will get chronic Kidney disease based on the following parameters age, bgr, bu, sc, sod, pot, hrmo, pcv, we, rc, sg, al, su, rbc, pc, pcc, ba, htn, dm, cad, appet, pe and ane.
Client has provided the dataset. (CKD.csv)

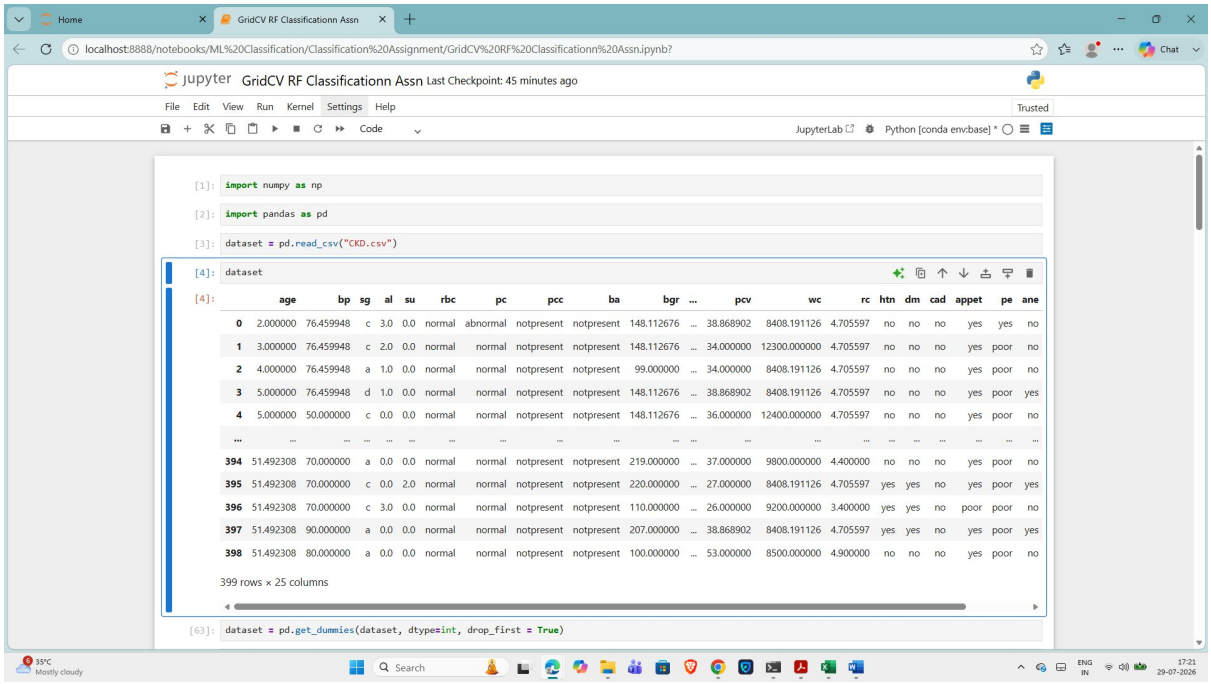
Domain and Learning Identification:
Requirement is clear with the client. Inputs will be age, bgr, bu, sc, sod, pot, hrmo, pcv, we, rc, sg, al, su, rbc, pc, pcc, ba, htn, dm, cad, appet, pe and ane. Output will be Classification.

Data preprocessing Details:
The following categorical columns need to be converted to Numerical. sg, al, rbc, pc, pcc, ba, htn, dm, cad, appet, pe, ane and Classification.

Nominal Method conversion for the following columns: sg, pc, rbc, pcc and ba
Ordinal Method conversion for the following columns: htn, dm, cad, appet, pe, ane and Classification

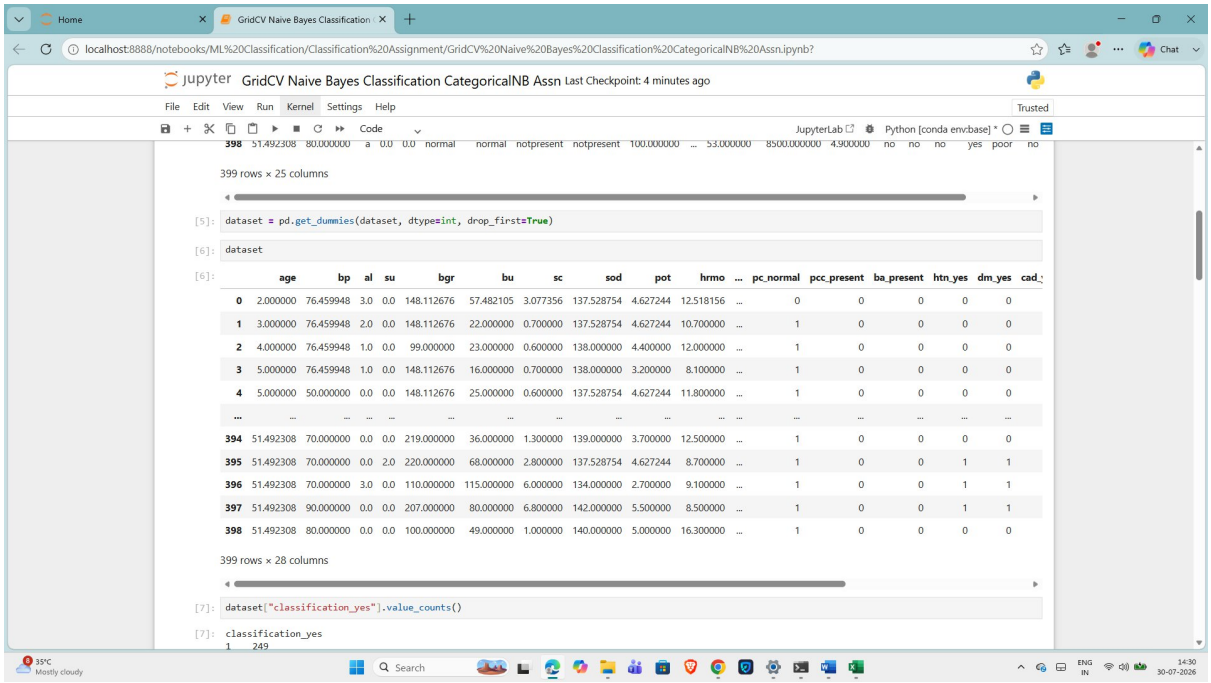
Domain and Learning Identification Result:
Since both inputs and output are numerical, the Domain would be 'Machine Learning'.
As client's requirements, inputs and output are clear, it falls under 'Supervised Learning'.
As output is Categorical (Yes or No), it falls under 'Classification'

2. Basic Information about the Dataset:



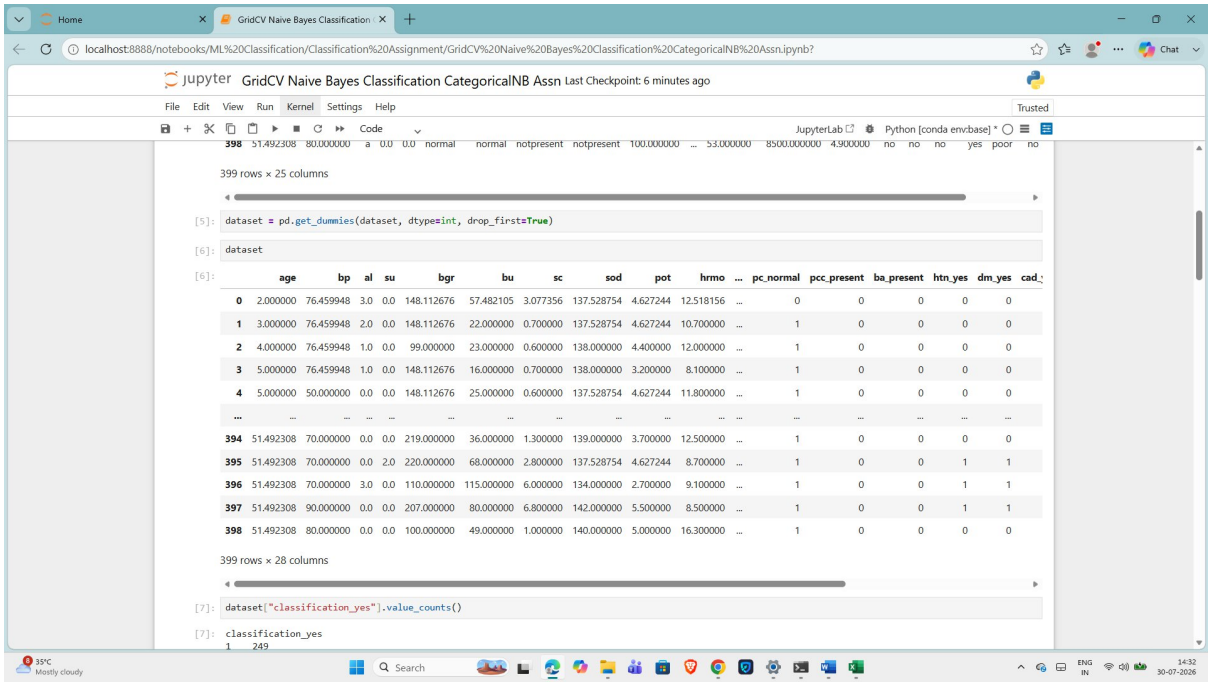
There are 399 rows and 25 columns.

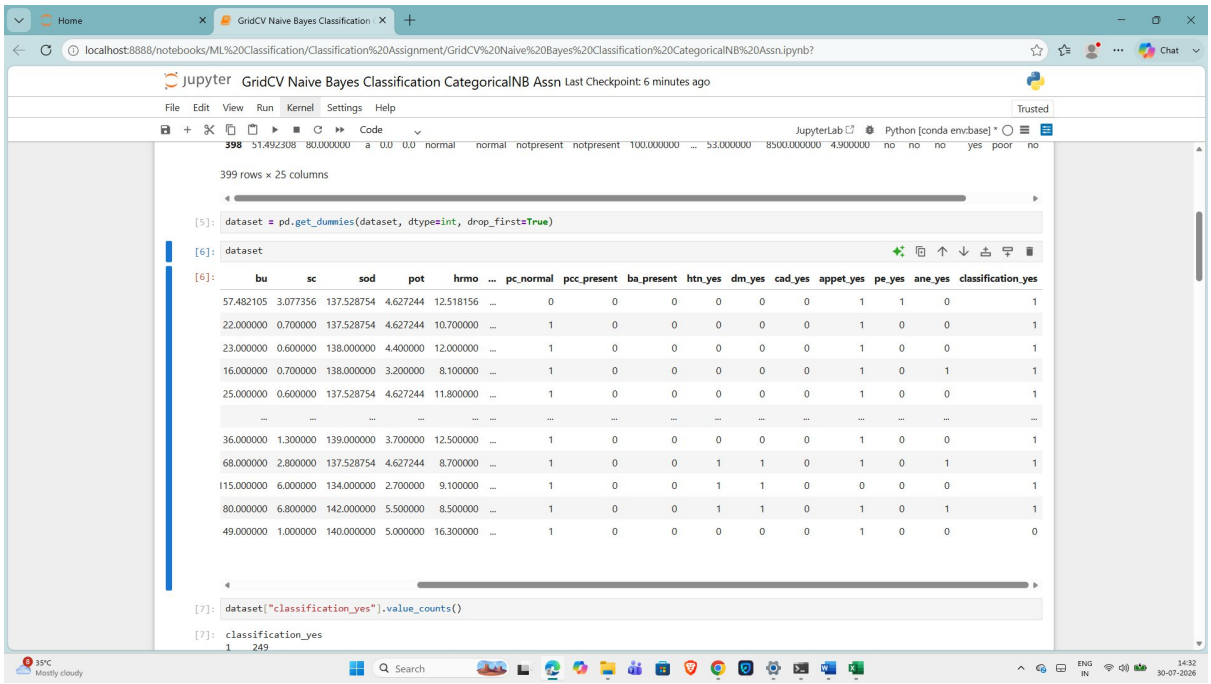
Post-Datapreprocessing, number of rows and columns are given in the below screenshot:



There are 399 rows and 28 columns.

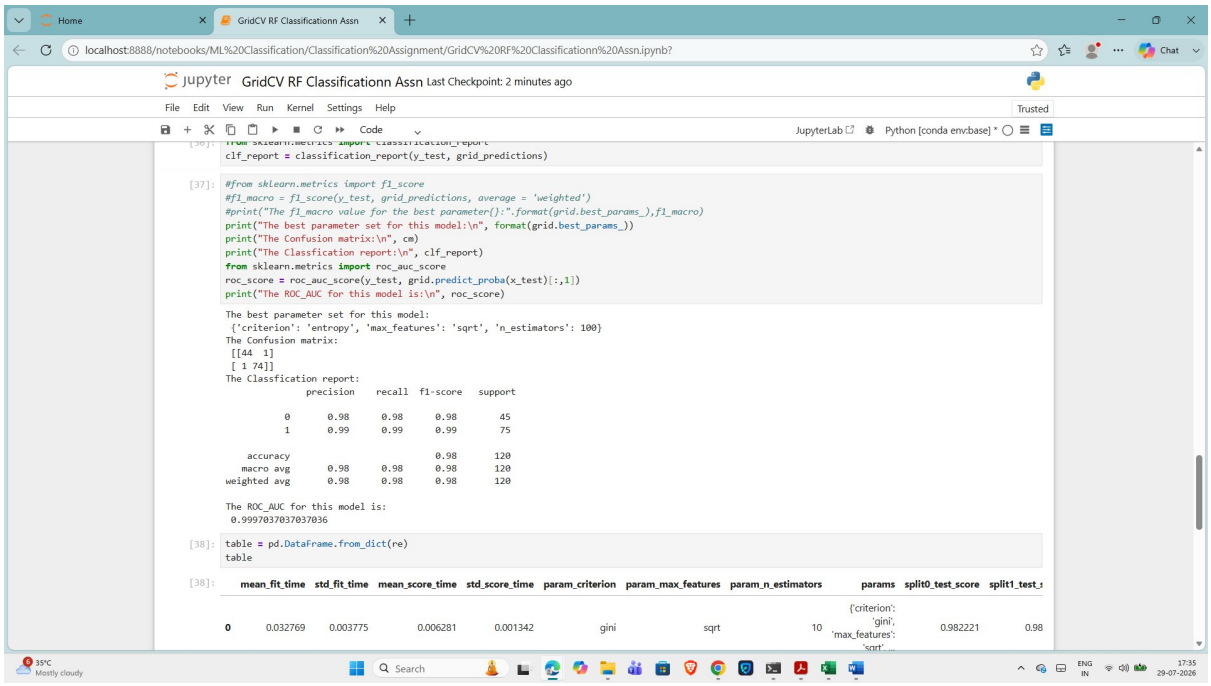
3. Data-preprocessing





4. Creating many models to find the best model for this dataset:

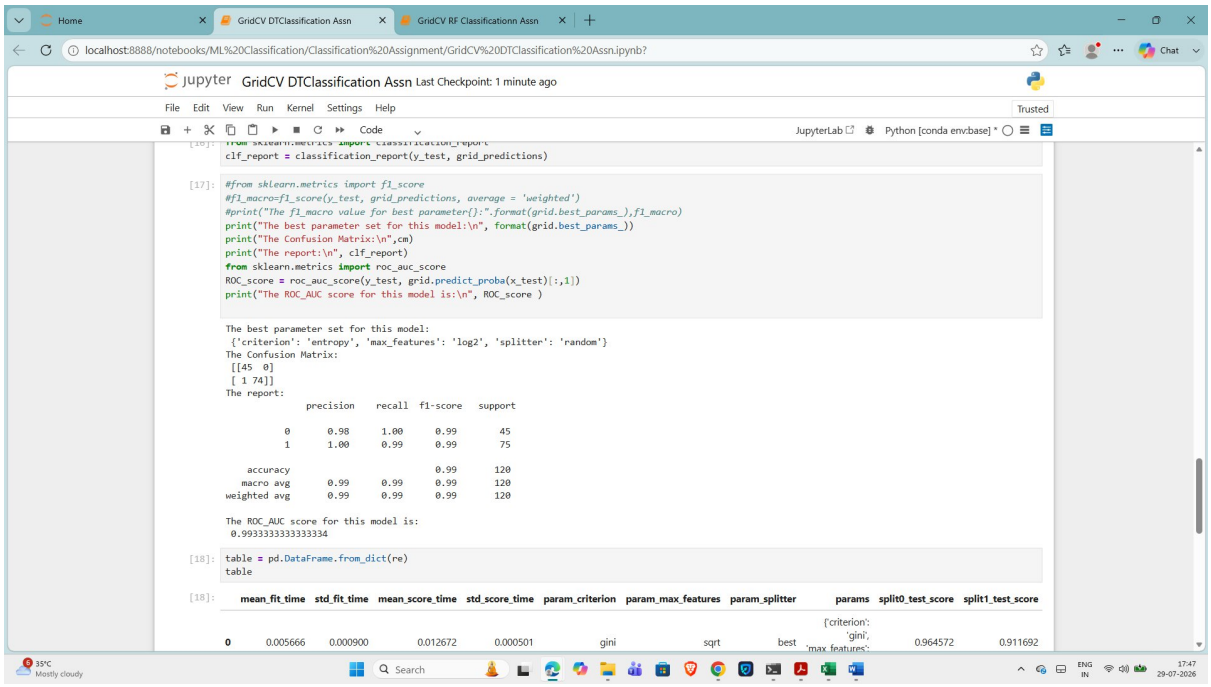
4.1 Grid CV Classification Random Forest – ROC-AUC Score:



ROC-AUC Score for RandomForest Classification(Grid):

0.9997037037037036

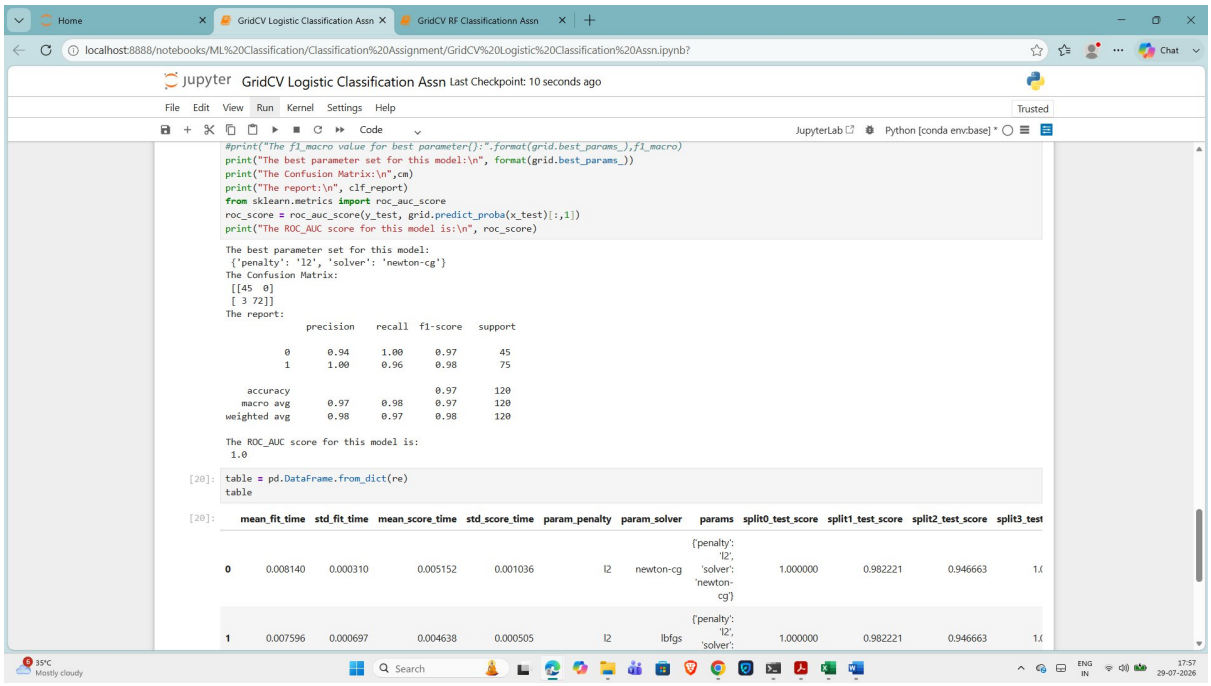
4.2 Grid CV Classification Decision Tree – ROC-AUC Score:



ROC_Score for DecisionTree Classification (Grid):

0.9933333333333334

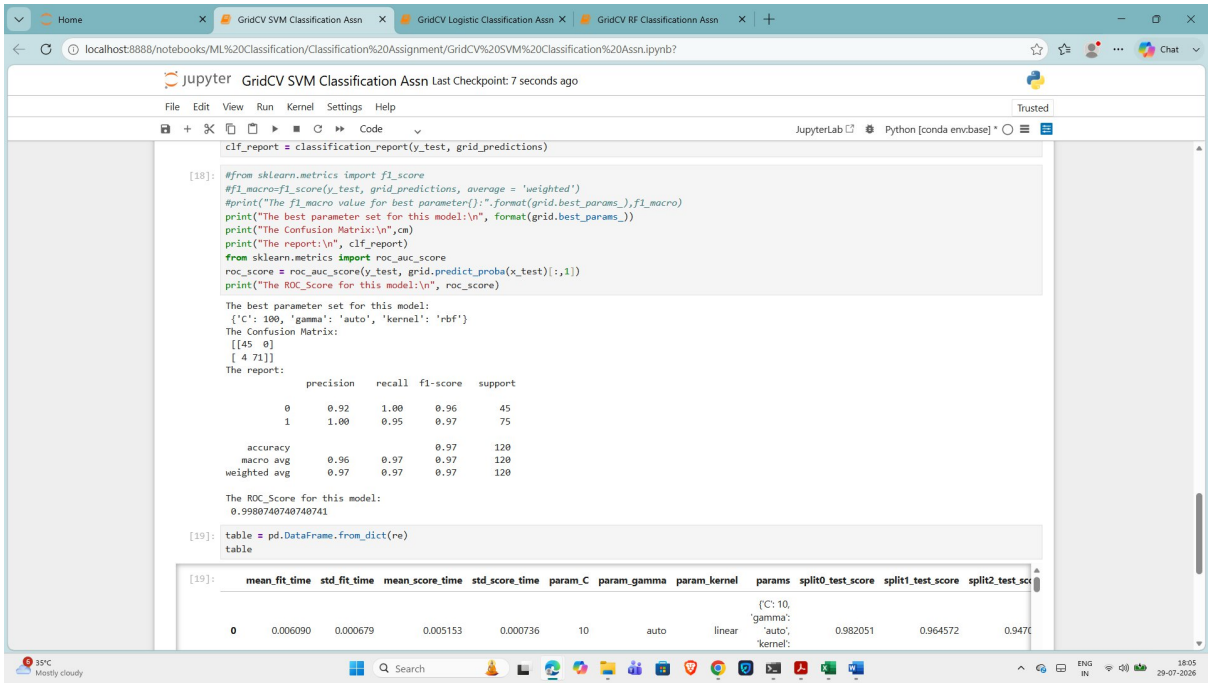
4.3 Grid CV Logistic Classification – ROC-AUC Score:



ROC_Score for Logistic Classification (Grid):

1.0

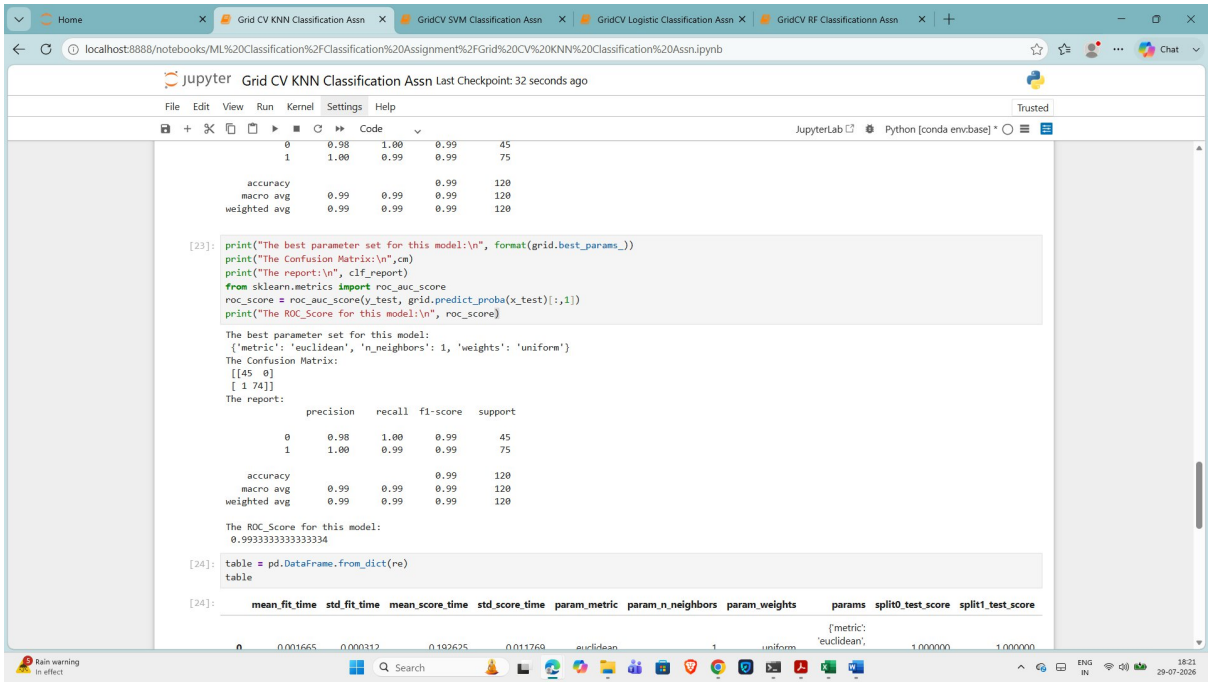
4.4 Grid CV SVM Classification – ROC-AUC Score:



ROC_Score for SVM Classification (Grid):

0.9980740740740741

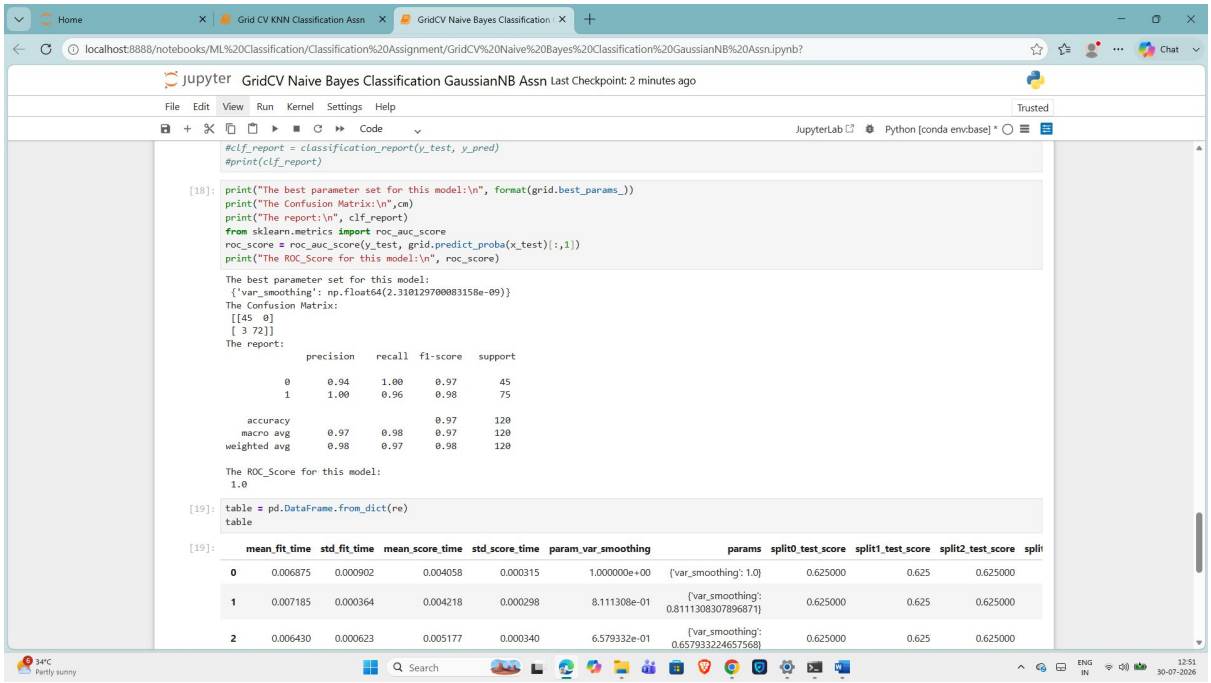
4.5 Grid CV KNN Classification – ROC AUC Score:



ROC Score for KNN Classification (Grid):

0.9933333333333334

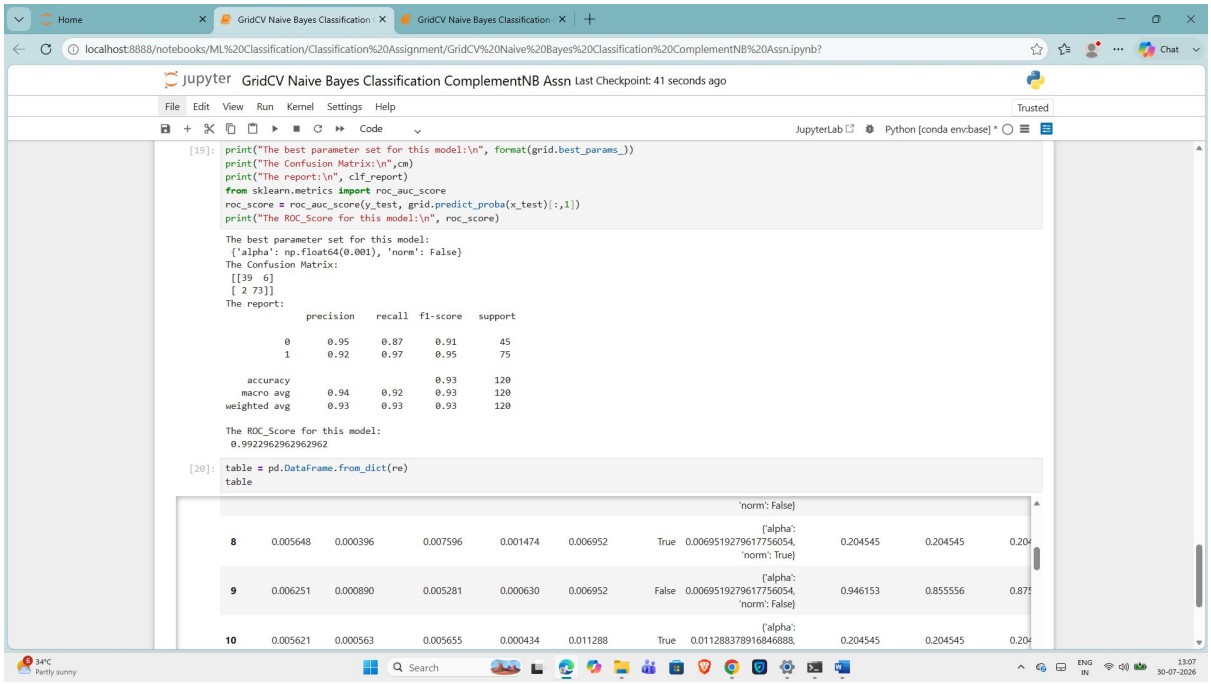
4.6 Grid CV GaussianNB Classification – ROC AUC Score:



ROC Score for GaussianNB Classification (Grid):

1.0

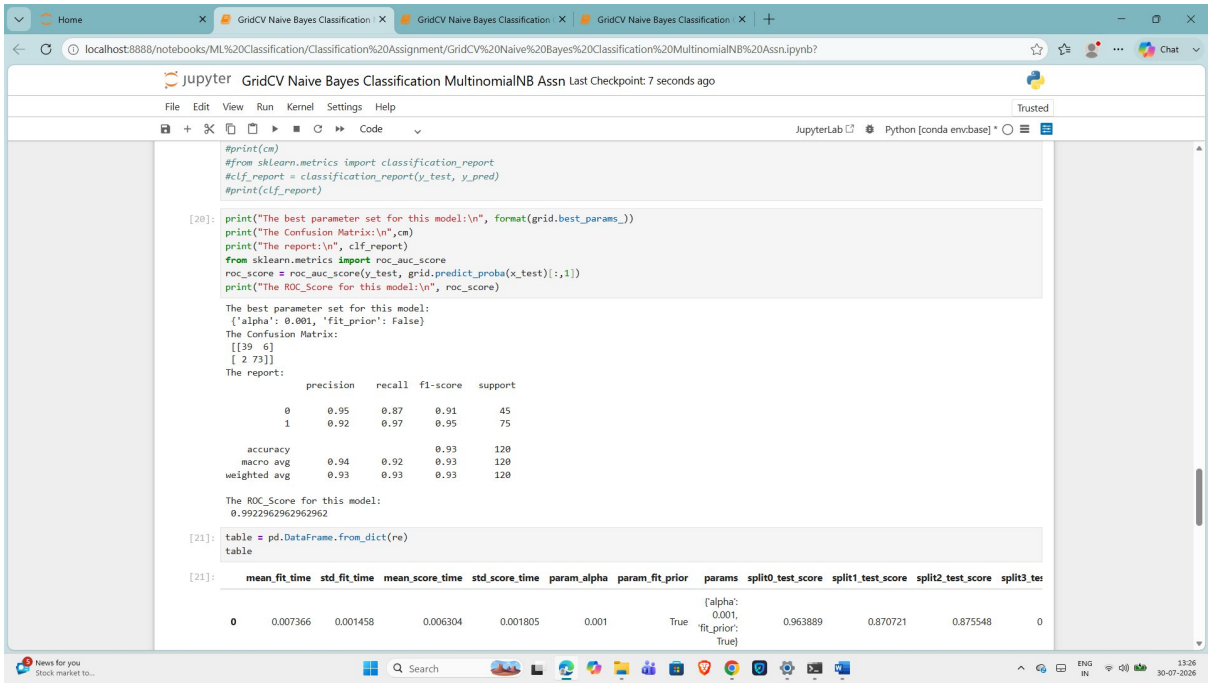
4.7 GridCV ComplementNB Classification – ROC-AUC Score:



ROC Score for ComplementNB Classification (Grid):

0.9922962962962962

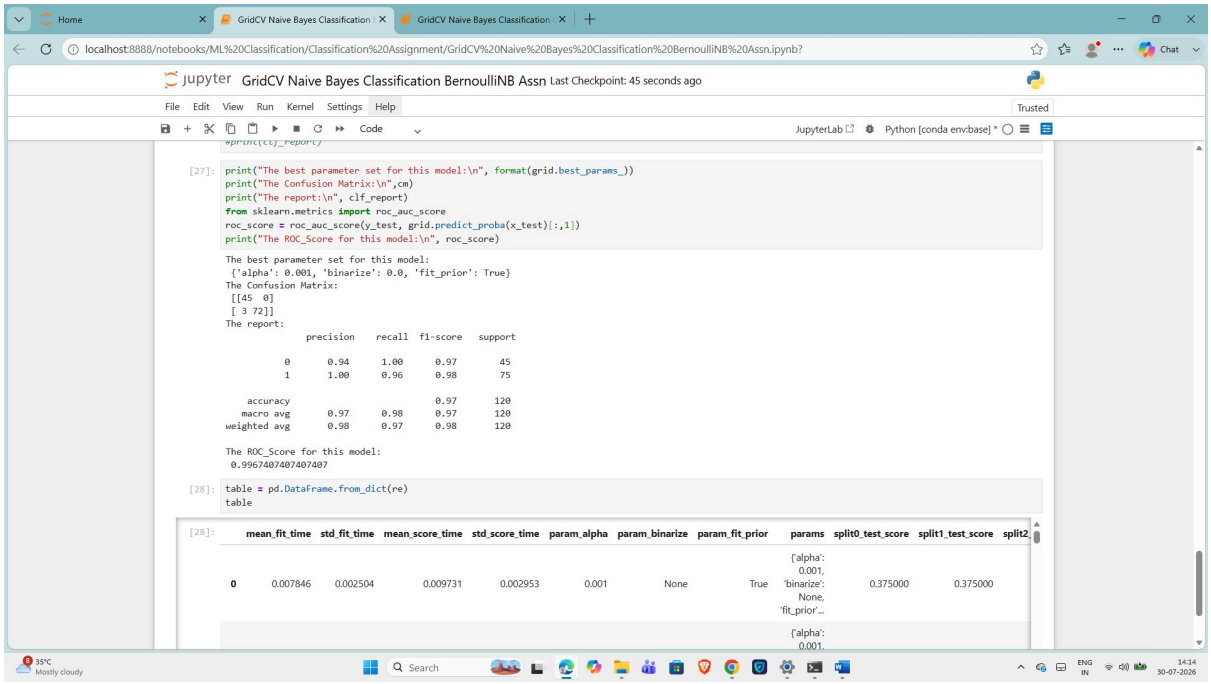
4.8 GridCV MultinomialNB Classification: ROC-AUC score



ROC-AUC Score for MultinomialNB Classification(Grid):

0.9922962962962962

4.9 GridCV BernoulliNB Classification: ROC_AUC Score:



ROC-AUC Score for BernoulliNB Classification(Grid):

0.9967407407407407

ROC_AUC Score for different Models:

S.No	Classification Algorithm	ROC_AUC Score	Ranking
1	RandomForest Classification Grid	0.9997037037037036	2
2	DecisionTree Classification Grid	0.9933333333333334	5
3	Logistic Classification Grid	1.0	1
4	KNN Classification Grid	0.9933333333333334	5
5	SVM Classification Grid	0.9980740740740741	3
6	GaussianNB Classification Grid	1.0	1
7	ComplementNB Classification Grid	0.9922962962962962	6
8	MultinomialNB Classification Grid	0.9922962962962962	6
9	BernoulliNB Classification Grid	0.9967407407407407	4

Two Best Models Listed below:

1.The best model for this dataset would be **'Logistic Regression Classification Grid'**, with highest **ROC_AUC score = 1**.

2.The best model for this dataset would be **'GaussianNB Classification Grid'**, with highest **ROC_AUC score = 1**.